HOMEWORK 13

Due date: Monday of Week 14

Exercises: 12.4, M.1, M.2, M.9, M.10, M.14.pages 75-77.

Exercises: 7.7, 7.8, 7.9, 7.10, 8.1, 8.2, 8.4, 11.1, 11.2, 11.3, 11.5, 11.8, M.7, pages 191-194.

Exercise: 5.7, page 223.

1. Cosets, double cosets, semi-direct product

Problem 1. *Let G be a group (not necessarily finite) and $H < G$ be a subgroup of G. Recall that G/H denotes the set of all left H-cosets and $H\backslash G$ denotes the set of all right H-cosets. Show that $f : G/H \to H\backslash G$ defined by $f(gH) = Hg^{-1}$ is well-defined and defines a bijection between G/H and $H\backslash G$. In particular, the number of left cosets is the same is the number of right cosets.*

There are at least two elements ϕ_0, ϕ_1 in $\mathrm{Aut}(\mathbb{Z}/n\mathbb{Z})$ defined by

$$\phi_0 = \mathrm{id}_{\mathbb{Z}/n\mathbb{Z}}; \phi_1(x) = -x, \forall x \in \mathbb{Z}/n\mathbb{Z}.$$

Consider the map

$$\phi : \mathbb{Z}/2\mathbb{Z} \to \mathrm{Aut}(\mathbb{Z}/n\mathbb{Z})$$

defined by $\phi(\bar{0}) = \phi_0, \phi(\bar{1}) = \phi_1$. It is clear that ϕ is a group homomorphism.

Problem 2. *Show that $\mathbb{Z}/n\mathbb{Z} \rtimes_\phi \mathbb{Z}/2\mathbb{Z}$ is isomorphic to D_n, the dihedral group of order $2n$.*

The next several problems are about double cosets, and most of them could be in last HW.

Problem 3. *Let F be a field and let $B_n(F) \subset \mathrm{GL}_n(F)$ be the upper triangular subgroup.*

(1) *Determine the double cosets $B_2(F)\backslash\mathrm{GL}_2(F)/B_2(F)$.*
(2) *How about $B_n(F)\backslash\mathrm{GL}_n(F)/B_n(F)$?*

This problem might be hard. It is related to the UPL (upper triangular, permutation subgroup, and lower triangular subgroup) decomposition of a matrix, See HW 3, Problem 5 of last year. If you don't know how to do the general problem, try the case when $n = 2$ and $F = \mathbb{F}_2$ (or $\mathbb{F}_3$).

Let $G \times X \to X$ be an action of a group G on a set X. Recall that $G\backslash X$ denote the set of orbits.

Problem 4. *Let G be a group and H, K are subgroups of G. Show the following basic properties of double cosets.*

(1) *For $x \in G$, the double coset HxK is a union of right H-cosets and a union of left K-cosets. More precisely,*

$$HxK = \coprod_{Hxk \in H\backslash HxK} Hxk = \coprod_{hxK \in HxK/K} hxK.$$

(2) *Let G act on the left cosets G/K from the left by $x.(gK) = (xg)K$. See Section 6.8 of Artin. We restrict this action to H and consider the action*

$$H \times G/K \to G/K$$

defined by $(h, gK) = (hg)K$. Show that there is a bijection between the double coset $H\backslash G/K$ and the set of orbits $H\backslash(G/K)$. This explains that the notation is consistent. There is a similar statement when we switch the role of H and K.

(3) *Suppose that all groups are finite. For $x \in G$, show that*

$$|HxK| = [H : H \cap xKx^{-1}]|K| = [K : K \cap x^{-1}Hx]|H|.$$

(4) *Show that*

$$[G:H] = \sum_{HxK \in H\backslash G/K} [K : K \cap x^{-1}Hx]$$

and

$$[G:K] = \sum_{HxK \in H\backslash G/K} [H : H \cap xKx^{-1}].$$

(5) *Consider the group action of* $(H \times K)$ *on* G *defined by*

$$((h,k),g) = hgk^{-1}, (h,k) \in H \times K, g \in G.$$

Check that this is a group action and there is a bijection between $H\backslash G/K$ *and the orbits of this action.*

(6) *Suppose* G *is finite. For* $h \in H, k \in K$*, consider* $G^{(h,k)} = \{g \in G : hgk = g\}$*. Show that*

$$|H\backslash G/K| = \frac{1}{|H||K|} \sum_{(h,k) \in H \times K} |G^{(h,k)}|.$$

For the last one, use Ex. M.7, page 194 of Artin. The other parts are routine.

2. Examples of group actions, regular polyhedral

Problem 5. *Determine the order of the group* $\mathrm{GL}_n(\mathbb{F}_p)$ *and* $\mathrm{Sp}_{2n}(\mathbb{F}_p)$*, where* p *is a prime number with* $p \neq 2$.

Hint: Consider the action of $\mathrm{GL}_n(\mathbb{F}_p)$ on $\mathbb{F}_p^n$ by left multiplication. Recall that $\mathrm{Sp}_{2n}(F)$ is the group $G = \{g \in \mathrm{GL}_{2n}(F) \mid g^t J_{2n} g = J_{2n}\}$, for a field F of odd characteristic. Here J_{2n} is any non-degenerate anti-symmetric matrix of rank $2n$. One can take

$$J_{2n} = \begin{pmatrix} & I_n \\ -I_n & \end{pmatrix},$$

or

$$J_{2n} = \begin{pmatrix} & & & & 1 \\ & & & 1 & \\ & & \dots & & \\ & -1 & & & \\ -1 & & & & \end{pmatrix},$$

where the number of $+1$ is n in the anti-diagonal.

Problem 6. *Let* $G = D_4$*, which is the symmetric group of a square. Let* X *be the set of two diagonals of this square. Analyze the action of* G *on* X *and describe the map* $D_4 \to Perm(X) \cong S_2$ *obtained from this action.*

Problem 7. *Let* T *be a tetrahedron and let* $G = G_T$ *be the symmetric group of* T.

(1) *Let* V *be the set of vertices of* T *and consider the action* $G \times V \to V$*. This will give a map* $G \to Perm(V) \cong S_4$*. From this to deduce that* $G \cong A_4$.
(2) *Let* X *be the set of the 3 line segments which connect the middle points of the opposite edges. Consider the action* $G \times X \to X$*. Consider the homomorphism* $G \to Perm(X) \cong S_3$ *obtained from this action. Show that this homomorphism is just composition* $A_4 \hookrightarrow S_4 \to S_3$ *(up to a relabeling of indices), where the map* $S_4 \to S_3$ *is the one in Example 2.5.13, page 50-51 of Artin's book.*

For part (1), use Exercise 5.7 of page 223.

Let $X_n = \{x_1, \dots, x_n\}$ be a set of n elements. Then S_n is just $Perm(X_n)$. For a subgroup $G \subset S_n$, we can consider the natural action of G on X_n. A subgroup of G is called **transitive**, if it acts transitively on X_n. Let $G = \{g \in S_n \mid gx_n = x_n\}$. Note that $G \cong S_{n-1}$, but G is not a transitive subgroup of S_n.

Problem 8. *Let* C *be a cube and let* $G = G_C$ *be the symmetric group of* C.

(1) *Let X be the set of the 4 diagonals of C. A diagonal of C is just a line segment which connects 2 opposite vertices. Consider the action $G \to X \to X$. Show that $G \cong S_4$.*
(2) *Using the first part to classify all transitive subgroups of S_4.*
(3) *Let F be the set of the 6 faces of C. Consider the action of G on F and the homomorphism $G \to Perm(F) = S_6$. Is this map injective? Is the image a transitive subgroup of S_6?*
(4) *Let Y be the set of the 3 pairs of opposite faces. Then $|Y| = 3$. Show that G acts on Y and the action gives the homomorphism $S_4 \to S_3$ in Example 2.5.13, page 50-51 of Artin's book.*
(5) *Call the 8 vertices as $v_1, \dots, v_8$ such that v_1, v_2, v_3, v_4 is the top square and v_5, v_6, v_7, v_8 is the bottom square and v_i is above v_{i+4}. (The picture is the same as what I drew in class. I will try to provide a picture if I have more time). Let T be the tetrahedron with vertices v_1, v_3, v_6, v_8. Describe the orbit and stabilizer of T under the natural action.*

For an answer of (2), see (16.9.1), page 493 of Artin's book. We will use this result later.

⟨problem-icosahedral⟩ **Problem 9.** *Let I be an icosahedral and let $G = G_I$ be the symmetric group of I. Then I has 12 vertices. Let X be the set of 6 main diagonals. Show that G acts on X and this gives a homomorphism $f : G \to Perm(X) = S_6$. Show that f is injective and its image is a transitive subgroup of S_6.*

We know that $G_I \cong A_5$ (this will be proved in next week, see page 199-200 of Artin's book) and thus S_6 contains a transitive subgroup which is isomorphic to A_5. On the contrary, if we view A_5 as a subgroup of S_6 by identifying it with as a subgroup of $H = \{h \in S_6 \mid hx_6 = x_6\}$, say, then it is not transitive. Recall that we know $\mathrm{Aut}(S_6)$ is strictly larger than S_6. In fact, we have $\mathrm{Out}(S_6) = C_2$. Thus there is a unique outer automorphism of S_6 (up to inner automorphisms). This unique outer automorphism is related to this transitive subgroup A_5 of S_6 as in Problem 9. The construction of such an outer automorphism is given in next homework.

We next give a different way to construct a transitive subgroup of S_6, which is isomorphic to A_5.

3. Projective space

Let F be a field and n be a positive integer. We introduce an equivalence relation on F^{n+1} as follows. For $\alpha, \beta \in F^{n+1}$, we say that $\alpha \sim \beta$ if there is an element $c \in F^\times$ such that $\beta = c\alpha$.

Problem 10. *Show that $R = \left\{(\alpha, \beta) \in F^{n+1} \times F^{n+1} : \alpha \sim \beta\right\}$ defines an equivalence relation on F^{n+1}.*

Denote $\mathbb{P}^n(F) := F^{n+1}/\sim$, which is the set of equivalence classes of the above equivalence relation. For example, $\mathbb{P}^1(F)$ is just the set of all lines which passes the origin. The set $\mathbb{P}^n(F)$ is called the n-dimensional projective space. Here we only view it as a set even it has more structures.

An element of $\mathbb{P}^n(F)$ is of the form $[\alpha]$, where $\alpha \in F^{n+1}$ and $[\alpha]$ denotes the equivalence class of α. Note that $[\alpha] = [\beta]$ if and only if $\beta = c\alpha$ for some $c \in F^\times$.

There is a group action of $\mathrm{GL}_{n+1}(F)$ on $\mathbb{P}^n(F)$ defined by

$$g.[\alpha] = [g\alpha], g \in \mathrm{GL}_{n+1}(F), \alpha \in F^{n+1}.$$

Consider the group $\mathrm{PGL}_{n+1}(F) = \mathrm{GL}_{n+1}(F)/Z$, where $Z = \{aI_{n+1}, a \in F^\times\}$ is the center of $\mathrm{GL}_{n+1}(F)$. The group $\mathrm{PGL}_{n+1}(F)$ is called the **projective general linear group** over F. An element of $\mathrm{PGL}_{n+1}(F)$ is of the form gZ, where $g \in \mathrm{GL}_{n+1}(F)$ and $gZ = hZ$ if and only if $g^{-1}h \in Z$, or $g = hz$ for some $z \in Z$. Similarly, we consider the group $\mathrm{PSL}_{n+1}(F) = \mathrm{SL}_{n+1}(F)/Z_0$, where $Z_0 = \left\{aI_{n+1} : a \in F^\times, a^{n+1} = 1\right\}$ is the center of $\mathrm{SL}_{n+1}(F)$.

Problem 11. *Show that the map $\mathrm{PGL}_{n+1}(F) \times \mathbb{P}^n(F) \to \mathbb{P}^n(F)$ defined by*

$$(gZ, [\alpha]) \mapsto [g\alpha]$$

is well-defined and it defines a group action of $\mathrm{PGL}_{n+1}(F)$ on $\mathbb{P}^n(F)$. Similarly, show that there is a group action of $\mathrm{PSL}_{n+1}(F)$ on $\mathbb{P}^n(F)$.

We next consider some very special cases.

Problem 12. *Let $\mathbb{F}$ be a finite field of order q. What is the cardinality of $\mathbb{P}^n(F)$? What is the cardinality of $\mathrm{PGL}_n(\mathbb{F})$? What is the cardinality of $\mathrm{PSL}_2(\mathbb{F})$?*

Here "cardinality" of a finite set means the number of elements in the finite set. For the second part and third part, think about how many elements are in the group $\mathrm{GL}_n(\mathbb{F})$ and $\mathrm{SL}_2(\mathbb{F})$.

Problem 13. *Let $\mathbb{F}_2$ be a finite field of 2 elements. Consider the action of $\mathrm{PSL}_2(\mathbb{F}_2)$ on $\mathbb{P}^1(\mathbb{F}_2)$. Show that*

$$\mathrm{PSL}_2(\mathbb{F}_2) \cong S_3.$$

Similarly, show that

$$\mathrm{PSL}_2(\mathbb{F}_3) \cong A_4.$$

Problem 14 (No need to submit a solution)**.** *Let $\mathbb{F}_4$ be a finite field of 4 elements. Consider the action of $\mathrm{PSL}_2(\mathbb{F}_4)$ on $\mathbb{P}^1(\mathbb{F}_4)$. Show that*

$$\mathrm{PGL}_2(\mathbb{F}_4) \cong A_5.$$

There is indeed a finite field of 4 elements. See HW 2 of last year. The above two problems are indeed Exercises 8.2, 8.3, page 287 of Artin's book. Hint: The following fact will be useful. If H is a subgroup of S_n of index 2, then $H = A_n$. This Exercise 5.7, page 223.

Problem 15 (No need to submit a solution)**.** *Consider the action of $\mathrm{PSL}_2(\mathbb{F}_5)$ on $\mathbb{P}^1(\mathbb{F}_5)$ and the homomorphism $\mathrm{PSL}_2(\mathbb{F}_5) \to S_6 = Perm(\mathbb{P}^1(\mathbb{F}_5))$ obtained from that. Show that this map is injective and its image is a subgroup is a transitive subgroup of S_6.*

One can show that $\mathrm{PSL}_2(\mathbb{F}_5) \cong A_5$ and the embedding $A_5 \to S_6$ obtained from the above map is roughly the same as the embedding one obtained from Problem 9. Here "roughly" means, one need to choose the indices carefully.